

CS 171 Homework 4

Merlin is a mobile phone app that is designed to help birders identify bird species that they encounter in the wild. One of the main features of the app allows users to upload a photograph of a bird they took in the wild and the app will identify the species based on the photograph. Merlin was developed as part of the Cornell Ornithology Lab and more information can be found here: <https://merlin.allaboutbirds.org/>. Luckily, being a science-focused group of bird enthusiasts, they have provided their entire database for researchers to use for free!

In this homework, you'll develop a Convolutional Neural Network to classify a subset of these images for eight common bird species in coastal California. The species and their images are summarized in the following table:

Species Number	Species	Number of Images
0	Pigeon Guillemot	81
1	California Gull	76
2	Brandt's Cormorant	66
3	Great Blue Heron	102
4	Great Egret	111
5	Turkey Vulture	108
6	Osprey	106
7	Barn Owl	109

The images for the birds above have been organized into 8 directories corresponding to their species and are provided with this assignment on Canvas. Since the focus of this assignment is on classification, the images have already been pre-processed to a common size of 200x200 pixels each.

As for all other assignments, you should restart your kernel and run all cells in succession to create a clean output in your notebook after you finish your assignment. Then, create a PDF of your notebook and turn both files in on Canvas.

To complete this assignment, create a clean Jupyter notebook with the following sections

- Problem 4.1: Organize the data
 - Data should be organized to facilitate training, testing, and validating a model
- Problem 4.2: Construct a CNN
 - Implement a CNN in PyTorch with appropriate layers and a loss function
- Problem 4.3: Training the CNN
 - Implement a training loop to optimize the weights in the network
- Problem 4.4: Assess the results of the CNN training
 - Plot model losses and compute other metrics for model performance. Based on performance, modify the model architecture and training to test different implementations of the model and improve the model results. Be sure to visualize some of the images and their true vs predicted labels in addition to the model metrics.
- Problem 4.5: Describe the training process and the performance of the final model
 - Write a paragraph describing your approach to constructing your CNN. Explain some of the issues that arose in building your network and some approaches you took to try to alleviate these issues. Comment on what your model may be learning and why some of mislabeled images may not be classified correctly by the model.

Note: You will not be graded on how well your final model performs. Instead, this focus of this assignment is on building a model, implementing steps to improve the model, and properly assessing the model based on various metrics of performance. This real-world example is more tricky than cartoon characters since many species of birds look similar. Good luck!